

# SIT315

## Comprehensive Interrupt – Based Sense-Think-Act System

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## Report and Reflection on Challenges and Key Findings

This project involved designing and testing an interrupt-driven Sense–Think–Act security system using an Arduino Uno in Tinkercad. The system used three pushbuttons to simulate an emergency button, a door sensor and a window sensor. A green LED represented normal operation, a red LED represented warning or alarm conditions, and a piezo buzzer sounded during an alarm. The main aim was to create a responsive system that could handle sensor events and a periodic timer task without blocking the program. This design meets the QP4 requirement of combining multiple sensors, pin-change interrupts and a timer-based periodic task.

The emergency button was connected to digital pin D2 and configured using `attachInterrupt()` with a falling-edge trigger. When the button was pressed, the interrupt service routine only set a volatile flag. The main loop then processed the flag, updated the system state and controlled the output devices. The door and window sensors were connected to D8 and D9. These pins belong to the same Port B pin-change interrupt group, so one shared ISR was used for both inputs. After an interrupt occurred, the main program read both pins and compared their new states with the previous states to identify which sensor had changed.

Timer1 was configured in CTC mode to generate an interrupt once every second. The timer ISR only set a timer flag, while the main loop used the flag to print a heartbeat message and blink the green LED when the system was safe. This kept the time-based task separate from the event-based sensor logic. No `delay()`, Serial output or complicated processing was placed inside the ISRs. This was important because the task requires short, non-blocking ISRs and clearly separated input, processing and actuation functions.

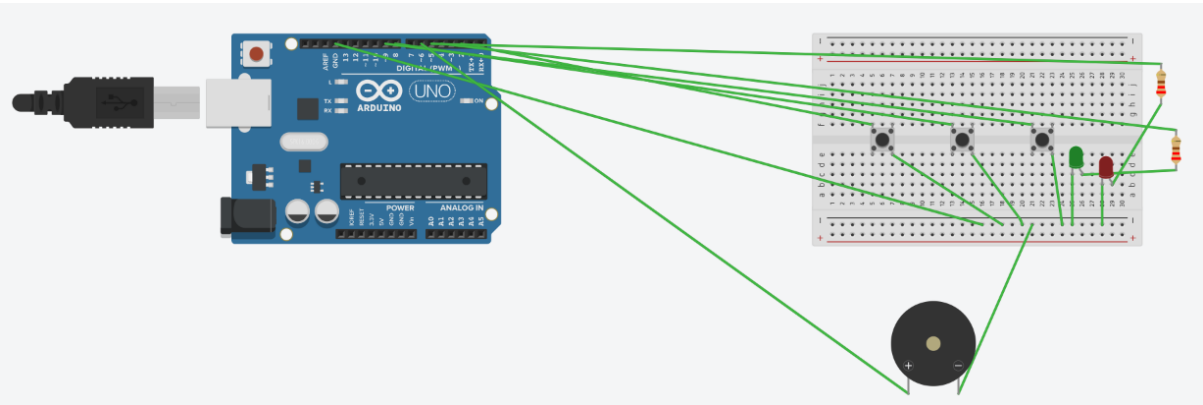
Four test conditions were completed. When no buttons were active, the system remained SAFE, and the green heartbeat LED blinked. When either the door or window sensor was active, the system entered WARNING, the red LED turned on and the buzzer remained off. When both the door and window sensors were active, the system entered ALARM, activating the red LED and buzzer. Pressing the emergency button also activated an immediate alarm through the external interrupt. Timer messages continued during all sensor tests, showing that periodic and event-driven operations worked together.

The main challenges were understanding the breadboard ground rail, connecting the LEDs and resistors correctly, and fixing a Tinkercad compilation error involving the `SystemState` type. I solved these issues by testing one component at a time, checking each connection and replacing the custom state type with simple numeric state constants. Debounce timing was also added to reduce repeated events caused by mechanical button bounce.

Overall, this task improved my understanding of external interrupts, pin-change interrupts, timer interrupts, volatile flags and safe ISR design. The most important finding was that interrupt routines should remain short, while decision-making, Serial output and actuator control should be completed in the main loop.

| TEST | INPUT CONDITION         | RESULT                                     |
|------|-------------------------|--------------------------------------------|
| 1    | No buttons active       | SAFE; green heartbeat LED blinked          |
| 2    | Door or window active   | WARNING; red LED on, buzzer off            |
| 3    | Door and window active  | ALARM; red LED and buzzer on               |
| 4    | Emergency button active | Immediate ALARM through external interrupt |

Tinkercad circuit diagram:



System monitoring log:

```
Serial Monitor

artbeat event
TIMER INTERRUPT: One-second heartbeat event
EXTERNAL INTERRUPT: Emergency button pressed
Emergency: ACTIVE | Door: Closed | Window: Closed | System: ALARM
Emergency button released
Emergency: Inactive | Door: Closed | Window: Closed | System: SAFE
TIMER INTERRUPT: One-second heartbeat event
TIMER INTERRUPT: One-second heartbeat event
PIN CHANGE INTERRUPT: Door sensor OPEN
Emergency: Inactive | Door: OPEN | Window: Closed | System: WARNING
TIMER INTERRUPT: One-second heartbeat event
PIN CHANGE INTERRUPT: window sensor OPEN
Emergency: Inactive | Door: OPEN | Window: OPEN | System: ALARM
TIMER INTERRUPT: One-second heartbeat event
TIMER INTERRUPT: One-second heartbeat event
TIMER INTERRUPT: One-second heartbeat event
PIN CHANGE INTERRUPT: Door sensor CLOSED
Emergency: Inactive | Door: Closed | Window: OPEN | System: WARNING
TIMER INTERRUPT: One-second heartbeat event
TIMER INTERRUPT: One-second heartbeat event
TIMER INTERRUPT: One-second heartbeat event
```

Private GitHub Repository:

<https://github.com/s225593298-AmizhthiniRajeshvarman/SIT315-TaskM1-Security-System>